

EduVision_DV: Higher Education Intelligence System

Final Project Documentation & Technical Report

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Executive Summary

Higher education institutions operate in an intensely competitive global environment where institutional ranking positions, research citation impact, faculty mentorship capacity, and student diversity directly determine funding allocations, student enrollment, and faculty recruitment. Historically, stakeholders have navigated disconnected datasets published across different ranking bodies (QS, Times Higher Education) and international organizations (The World Bank). In this project, I engineered **EduVision_DV**, an audit-compliant educational analytics platform and an integrated suite of four Tableau dashboards. The pipeline extracts, normalizes, and reconciles multi-source datasets covering 1,503 institutions across 106 countries, engineers six defensible Key Performance Indicators (KPIs), and achieves a 100% verification pass rate across all relational and mathematical test suites.

1. Dataset Acquisition & Provenance

Four specialized datasets were acquired to evaluate distinct institutional and macroeconomic dimensions:

Dataset Title	Publisher	Coverage	Role in Architecture
QS World Rankings 2025	Quacquarelli Symonds	1,503 universities, 106 nations	Primary institutional dimension ('dim_university'), global ranks, overall score, academic reputation.
THE World Rankings 2024	Times Higher Education	2,673 institutions globally	Research environment, normalized citations, teaching score, industry income ('fact_research').
World University Rankings 2023	Times Higher Education	2,341 universities	Student demographics: FTE student headcount, students per staff, international % ('fact_student').
World Bank EdStats	The World Bank Group	242 countries (2010–2023)	Macroeconomic indicators: public education spend (% GDP), tertiary enrollment ('fact_country_education').

University Match Rate vs. Data Completeness: Adhering strictly to Section 7 of the project guidelines, university match rate (natural 55–60% overlap between distinct ranking bodies) was not artificially inflated through aggressive fuzzy matching. A false match is significantly worse than a missing match. Data completeness on required dimensional keys achieved 100%.

2. Data Cleaning & Normalization Methodology

Data cleaning was executed in Python using Pandas following a deterministic 5-stage algorithm:

- Header Sanitization:** Converted column names to lowercase snake_case, stripped UTF-8 BOM marks, and removed special characters.
- Acronym & Punctuation Stripping:** Applied regex `r'\(.*?\)'` to remove parenthetical abbreviations (e.g. '(MIT)'), and `r'[^\w\s]'` to eliminate punctuation, followed by whitespace compression `r'\s+'`.
- Deterministic Entity Resolution:** Merged on normalized canonical names, yielding **849 exact research matches** and **804 student matches** with zero false positives. Eliminated dangerous row-order fallback indexing.
- Geopolitical Harmonization:** Mapped country variations ('USA', 'US' -> 'United States'; 'UK' -> 'United Kingdom') and assigned structured surrogate keys ('C0001'..'C0106').
- Missing Value Integrity:** Enforced Category A/B/C rules. Preserved 'NaN' on non-evaluated metrics to prevent skewing statistical averages. Calculated inverse percentile ranks for institutions outside the top 500.

3. Relational Star Schema Architecture

To eliminate join multiplication and ensure sub-second Tableau dashboard filtering, a Star Schema was deployed:

Table Name	Table Type	Primary / Foreign Key	Row Count & Grain
`dim_university`	Conformed Dimension	PK: `university_id` FK: `country_id`	1,503 rows (1 per institution)
`dim_country`	Conformed Dimension	PK: `country_id`	106 rows (1 per sovereign nation)
`fact_university_performance`	Fact Table	FK: `university_id`	1,503 rows (QS rankings & reputation)
`fact_research`	Fact Table	FK: `university_id`	849 rows (THE research & citations)
`fact_student`	Fact Table	FK: `university_id`	804 rows (WUR student demographics)
`fact_country_education`	Fact Table	FK: `country_id`	2,243 rows (World Bank macro stats)
`kpi_master`	Analytical Master	PK: `university_id`	1,503 rows (Pre-aggregated master)

4. Higher Education Key Performance Indicators (KPIs)

All six KPIs were engineered following transparent mathematical formulations, avoiding arbitrary field substitutions:

KPI # & Name	Primary Source	Mathematical Derivation / Formula	Unit & Target Scale
KPI 1: Global Ranking Score	QS Overall Score	QS Overall Score for top 500; inverse percentile rank formula imputed for rank > 500.	0.0 – 100.0 Score
KPI 2: Research Impact Score	QS / THE Citations	QS Citations per Faculty Score normalized for faculty size.	0.0 – 100.0 Score
KPI 3: Faculty-to-Student Ratio	WUR 2023	Actual empirical ratio: Students per academic staff member.	Ratio (e.g. 8.2 : 1)
KPI 4: International Student %	WUR 2023	True empirical percentage: (International FTE / Total FTE) * 100.	0.0% – 100.0%
KPI 5: Academic Reputation Score	QS Academic Rep	Global survey of over 100,000 active academics worldwide.	0.0 – 100.0 Score
KPI 6: Research Productivity Index	Derived Composite	0.50 * Research Score + 0.30 * Citation Score + 0.20 * International Research Network.	0.0 – 100.0 Index

Calculated Values for World-Class Institutions:

Institution	Rank	Global Score	Res Impact	Staff Ratio	Intl %	Acad Rep	Prod Index
Massachusetts Inst of Tech (MIT)	1	100.00	100.00	8.2 : 1	33.0%	100.00	97.30
Imperial College London	2	98.50	93.90	11.2 : 1	61.0%	98.50	95.40
University of Oxford	3	96.90	84.80	10.6 : 1	42.0%	100.00	95.44
Harvard University	4	96.80	100.00	9.6 : 1	25.0%	100.00	99.87
University of Cambridge	5	96.70	84.60	11.3 : 1	39.0%	100.00	95.24

5. Tableau Dashboard Suite & Interactivity

The unified workbook (`EduVision\_DV.twbx`) contains four interconnected dashboards styled in dark slate (`#0F172A`) with high-contrast accents:

- **Dashboard 1 (University Overview):** Landing view featuring 6 top KPI cards, ranked Top 10 bar chart, Reputation vs. Score scatter, and geographic map.
- **Dashboard 2 (Research Analytics):** Explores Research Productivity Index rankings, citation velocity, and cross-border research networks.
- **Dashboard 3 (Student Analytics):** Analyzes teaching capacity (student-to-staff ratios), international student %, and gender diversity.
- **Dashboard 4 (Country Comparison):** Macroeconomic education benchmarking using World Bank EdStats (national capacity, spend % GDP, tertiary enrollment).

Interlinking Action Architecture: Clicking any university on Dashboard 1 triggers a filter action passing `university\_id` to Dashboards 2 and 3. Clicking a nation passes `country\_id` to Dashboard 4. A persistent top navigation bar enables direct transitions with persistent filters.

6. Quality Assurance & Testing Results

An automated test suite (``validate_and_test.py``) verified all components against 12 formal test cases with a **100% pass rate**:

- **Primary Key Uniqueness:** 1,503 / 1,503 unique keys in ``dim_university``; 106 / 106 unique keys in ``dim_country``.
- **Referential Integrity:** 100% of rows in fact tables match valid dimension keys with **0 orphan records**.
- **KPI Boundaries:** All scores lie strictly within [0.0, 100.0]; all faculty ratios are positive real numbers; international % is within [0.0%, 100.0%].
- **Completeness:** 0.0% missing on critical primary identifiers.

Submission Sign-Off: Verified, audit-compliant, and fully portfolio-ready.

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